

Theorems:

Week 1: Theorem 1.1 Let A and B be statements. Then $A \Rightarrow B$ is equivalent to $\overline{A} \vee B$. That is, $(A \Rightarrow B) \Leftrightarrow (\overline{A} \vee B)$.

Theorem 1.2 Let A and B be statements. Then $A \Rightarrow B$ is equivalent to $\overline{B} \Rightarrow \overline{A}$. That is, $(A \Rightarrow B) \Leftrightarrow (\overline{A} \vee B)$.

Theorem 1.3 Let A and B be statements. Then $\overline{A \vee B}$ is equivalent to $\overline{A} \wedge \overline{B}$.

Theorem 2.4 Given any two real numbers $a < b$, there is some $r \in \mathbb{Q}$ such that $a < r < b$.

Theorem 2.5 The number $\sqrt{2}$ is irrational.

Week 2:

Theorem 3.7 -AM-GM Inequality Let $n \in \mathbb{N}$ and $x_1, \dots, x_n \geq 0$. Then $(x_1 \cdot x_2 \dots x_n)^{1/n} \leq \frac{(x_1 + \dots + x_n)}{n}$

Theorem 3.8 -Cauchy-Schwarz Inequality Let $n \in \mathbb{N}$, $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}$. Then $x_1 y_1 + \dots + x_n y_n \leq$

$$\sqrt{x_1^2 + \dots + x_n^2} \sqrt{y_1^2 + \dots + y_n^2}$$

Theorem 3.8 Triangle Inequality For $x, y \in \mathbb{R}$, $|x + y| \leq |x| + |y|$.

Corollary 3.12 Reverse Triangle Inequality $||x| - |y|| \leq |x + y|$. in particular: $|x| - |y| \leq |x + y|$

Theorem 4.1 -Geometric Series Let $x \neq 1$ be a real number. Then for $N \in \mathbb{N}$, $\sum_{n=1}^N x^n = \frac{x-x^{N+1}}{1-x}$ and $\sum_{n=0}^N x^n = \frac{1-x^{N+1}}{1-x}$

Theorem 4.2 Every Integer $n \geq 2$ can be written as a product of primes $n = p_1 \dots p_k$

Week 3: Completeness Axiom for the real numbers: Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Theorem 5.1 the Archimedean property holds. That is, for any $x, y \in \mathbb{R}$ with $x, y > 0$, there is some $n \in \mathbb{N}$ such that $ny > x$.

Theorem 6.6 Let $x > 0$ and $k \in \mathbb{N}$. Then there is a unique $y > 0$ such that $y^k = x$.

Week 4: Theorem 7.1 Monotone Convergence Theorem Let (a_n) be an increasing sequence of real numbers that is bounded above (i.e. there is M so that $a_n \leq M$ for all n). Then (a_n) converges to some limit. If (a_n) is a decreasing sequence and is bounded below, then (a_n) converges to some limit.

Theorem 8.1 Given a sequence (a_n) with $a_n \in 0, 1, \dots, 9$ for all n , the decimal expansion $x = 0.a_1 a_2 \dots a_n \dots$ given by the limit of the sequence $x_n = \frac{a_1}{10} + \frac{a_2}{100} + \dots + \frac{a_n}{10^n}$ defines a real number x satisfying $0 \leq x \leq 1$

Theorem 8.2 Let $x \in \mathbb{R}$ satisfy $0 \leq x < 1$. Then there is a sequence of integers (a_n) , with $a_n \in 0, 1, \dots, 9$ for all n , so that if we let $x_n = \frac{a_1}{10} + \frac{a_2}{100} + \dots + \frac{a_n}{10^n}$ then $x_n \rightarrow x$ as $n \rightarrow \infty$

Theorem 8.3 Let $0 \leq x < 1$ and suppose that $x = 0.a_1 a_2 \dots = 0.b_1 b_2 \dots$. Let ℓ be the smallest integer for which $a_\ell \neq b_\ell$ and suppose that $a_\ell < b_\ell$. Then $b_\ell = a_\ell + 1$, and for all $k > \ell$ we have $b_k = 0$ and $a_k = 9$

Lemma 8.4 if $x \geq 0$ is rational, then it has a periodic decimal expansion, that is, $x = a_0.a_1 a_2 \dots \overline{a_k b_1 b_2 \dots b_\ell}$

Corollary 8.5 If p and q are positive integers, the rational number p/q has a decimal expansion with period at most q .

Lemma 8.6 If $x \geq 0$ has a periodic decimal expansion, then x is rational.

Theorem 8.7 A number $x \geq 0$ is rational $\Leftrightarrow$ it has a periodic decimal expansion.

Theorem 8.11 -Comparison Test If $b_j \geq 0$ for all j and $\sum_{j=1}^{\infty} b_j$ is convergent, and if $|a_j| \leq b_j$ for all j , then $\sum_{j=1}^{\infty} a_j$ is also convergent, and moreover $|\sum_{j=1}^{\infty} a_j| \leq \sum_{j=1}^{\infty} b_j$

Week 5: Lemma 9.1 For all $u, v \in \mathbb{C}$: (1) $\overline{u+v} = \overline{u} + \overline{v}$ (2) $\overline{uv} = \overline{u} * \overline{v}$ (3) $|u+v| = |u| + |v|$

Theorem 9.2 -De Moivre's If we let $z = r(\cos\theta + i\sin\theta)$, and $n \in \mathbb{N}$ then $z^n = r^n(\cos n\theta + i\sin n\theta)$ $z^{-n} = r^{-n}(\cos -n\theta + i\sin -n\theta) = \frac{1}{r^{-n}}(\cos(n\theta) - i\sin(n\theta))$

Theorem 9.3 -Eulers Formula $e^{i\theta} = \cos(\theta) + i\sin(\theta)$.

Theorem 9.4 -Roots of Unity The solutions to $z^n = 1$ are $1, w, \dots, w^{n-1}$ where $w = e^{\frac{2\pi i}{n}}$ That is, they are $e^{\frac{2\pi ki}{n}}$ for $k = 0, 1, \dots, n-1$.

Theorem 10.1 -Abel-Ruffini There is no formula (like the quadratic formula) for the roots of a polynomial of degree ≥ 5

Theorem 10.2 -Fundamental Theorem of Algebra Any complex polynomial has at least one root in $\mathbb{C}$.

Theorem 10.3 -Factorization Theorem If p is a degree n polynomial, then there are n roots $r_1, \dots, r_n \in \mathbb{C}$ and a number $a \in \mathbb{C}$ so that $p(z) = a(z - r_1)(z - r_2) \dots (z - r_n)$.

Theorem 10.4 -Real Polynomials have conjugate roots If $p(x)$ has *real* coefficients and r is a root, so is $\bar{r}$.

Theorem 10.5 -Root-Coefficient Theorem $p(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$, has roots $r_1, \dots, r_n$ (counting multiplicities), then $r_1 + \dots + r_n = -a_{n-1}$ $r_1 \dots r_n = (-1)^n a_0$. In general, if s_j denotes the sum of all products of j -tuples of the roots (e.g. $s_2 = r_1 r_2 + r_1 r_3 + r_2 r_3 + \dots$), then $s_j = (-1)^j a_{n-j}$.

Week 6: Theorem 11.1 - The Remainder Theorem Let $a \in \mathbb{N}$ and $b \in \mathbb{Z}$. There are unique integers $q \in \mathbb{Z}$ and $0 \leq r < a$ such that $b = qa + r$

Lemma 11.2 If a divides b and b divides a then $a = \pm b$

Lemma 11.3 -Linear Combination Lemma Let a, b , and d be integers. If d divides both a and b , then d divides every integer linear combination of a and b . In other words, for every pair of integers $m, n \in \mathbb{Z}$, the integer d also divides $ma + nb$.

Theorem 11.4 -Bezout's Identity Let a and b be integers. Then $\gcd(a, b)$ is an integer linear combination of a and b . That is, there are integers s and t such that $\gcd(a, b) = sa + tb$.

Corollary 11.5 Let $a, b, c \in \mathbb{Z}$. Suppose $c \neq 0$, c divides ab and $\gcd(a, c) = 1$. Then c divides b .

Corollary 11.6 Let $a, b \in \mathbb{Z}$ and let $d \in \mathbb{Z}$ be such that d divides a and d divides b . Then d divides $\gcd(a, b)$.

Corollary 11.7 Let d be a common divisor of a and b , which is divisible by all divisors of a and b . Then $d = \pm \gcd(a, b)$.

Corollary 11.8 If $a, b \in \mathbb{Z}$, p is prime, and p divides ab , then either p divides a or p divides b (or both).

Corollary 11.9 If $n = p_1^{m_1} \dots p_k^{m_k}$, where each of $p_1, \dots, p_k$ is prime, and if p is a prime number that divides n , then $p = p_i$ for some $i = 1, \dots, k$.

Theorem 11.10 -Fundamental Theorem of Arithmetic Let $n \geq 2$ be an integer. *Existence* Then n is equal to a product $n = p_1^{r_1} \dots p_k^{r_k}$ of powers of prime numbers, where $p_1 < \dots < p_k$ and $r_i \geq 0$ for all i . (Uniqueness) The factorization is unique. That is, if we also have $p_1^{r_1} \dots p_k^{r_k} = n = q_1^{s_1} \dots q_\ell^{s_\ell}$ where $q_1 < \dots < q_\ell$ are primes and $s_i > 0$, then $k = \ell$, $p_i = q_i$, and $r_i = s_i$ for all i .

Lemma 11.11 -Existence of prime decomposition, Part 1. Every integer $n \geq 2$ can be written as a product of primes $n = p_1 \dots p_k$.

Lemma 11.12 -Existence of prime decomposition, Part 2. Every integer $n \geq 2$ can be written as a product of powers of primes $n = p_1^{r_1} \dots p_k^{r_k}$ where $p_1 < \dots < p_k$ and $r_i \geq 0$.

Lemma 11.13 -Uniqueness of Prime decomposition. Every integer n can be written as a *unique* product of powers of primes $n = p_1^{r_1} \dots p_k^{r_k}$

Theorem 11.14 Let $n = p_1^{a_1} \dots p_k^{a_k}$ be a prime decomposition. Then m divides n if and only if $m = p_1^{b_1} \dots p_k^{b_k}$ with each $0 \leq b_i \leq a_i$

Theorem 11.15 Let a and b have prime factorizations, $a = p_1^{r_1} \dots p_m^{r_m}$, $b = p_1^{s_1} \dots p_m^{s_m}$. Here p_i 's are distinct, but r_i and s_i are allowed to be zero. Then: (1) $\gcd(a,b) = p_1^{\min(r_1,s_1)} \dots p_m^{\min(r_m,s_m)}$ (2) $\text{lcm}(a,b) = p_1^{\max(r_1,s_1)} \dots p_m^{\max(r_m,s_m)}$ (3) $\text{lcm}(a,b) = ab/\gcd(a,b)$.

Lemma 11.16 Let $r \in \mathbb{Q}$. Then there exist unique coprime integers a and b such that $r = \frac{a}{b}$ and $b \geq 1$.

Theorem 11.17 Let n be a positive integer. Then $\sqrt{n} \in \mathbb{Q}$ if and only if n is a perfect square, that is, $n = m^2$ for some integer m .

Theorem 11.18 If $a, b \in \mathbb{N}$ are coprime, and ab is an n th power, then so are a and b .

Week 7: Theorem 12.1 Let $m \in \mathbb{N}$ and $a, b \in \mathbb{Z}$. The following are equivalent: (1) $a \equiv b \pmod{m}$; (2) m divides $a - b$; (3) a and b have same remainder when divided by m ; (4) $b = qm + a$ for some $q \in \mathbb{Z}$.

Theorem 12.2 We have the following properties for integers $a, b, c \in \mathbb{Z}$ and $m \in \mathbb{N}$: (1) $a \equiv a \pmod{m}$; (2) If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$. (3) If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.

Theorem 12.3 Suppose $a \equiv x \pmod{m}$ and $b \equiv y \pmod{m}$. Then (1) $-a \equiv -x \pmod{m}$; (2) $a + b \equiv x + y \pmod{m}$; (3) $ab \equiv xy \pmod{m}$; (4) for any natural number k , one has $a^k \equiv x^k \pmod{m}$.

Theorem 12.4 Let a and m be coprime. If $x, y \in \mathbb{Z}$ and $xa \equiv ya \pmod{m}$, then $x \equiv y \pmod{m}$. In particular, if p is prime and p does not divide a , then $xa \equiv ya \pmod{p}$ implies that $x \equiv y \pmod{p}$.

Theorem 12.5 The equation $ax \equiv b \pmod{m}$ has a solution if and only if $\gcd(a, m)$ divides b .

Theorem 12.6 Let $m \geq 2$ be a natural number. Then $\mathbb{Z}/m$ is a field if and only if m is prime.

Theorem 12.7 -Fermat's Little Theorem Let a be an integer, and let p be a prime number. If p does not divide a , then $a^{p-1} \equiv 1 \pmod{p}$

Theorem 12.8 -Fermat's Little Theorem Let $n \in \mathbb{N}$ and p be a prime number. Assume n and $p - 1$ are coprime and p does not divide b . Then the equation $x^n \equiv b \pmod{p}$ has exactly one solution in $x \in 1, \dots, p - 1$

Week 8: Theorem 13.1 Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions. (1) If f and g are injective, so is $g \circ f$. (2) If f and g are surjective, so is $g \circ f$. (3) If f and g are bijective, so is $g \circ f$.

Theorem 13.2 Let $f: X \rightarrow Y$ be a function and $B, B' \subseteq Y$. Then $f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$ and $f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$

Theorem 13.3 Let X and Y be finite sets and let $f: X \rightarrow Y$ be a function. (1) If f is injective, then $|X| \leq |Y|$. (2) If f is surjective, then $|X| \geq |Y|$. (3) If f is bijective, then $|X| = |Y|$.

Theorem 14.1 $\mathbb{Q} \cong \mathbb{N}$.

Theorem 14.2 Let S be a set and let $P(S)$ denote its power set, consisting of the subsets of S , i.e. $P(S) = \{A : A \subseteq S\}$. Then $|S| < |P(S)|$.

Week 9: Theorem 15.1 -Multiplication Principle Let P be a process consisting of n stages such that at each stage i , there are a_i choices we can make, and no two distinct choices can result in the same outcome. Then after n stages, the total number of outcomes is $a_1 a_2 \dots a_n$.

Theorem 15.3 -Ordering Theorem Given a set S of n distinct objects, there are $n!$ ways of ordering them.

Theorem 15.5 For integers $0 \leq k \leq n$, we have $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Corollary 15.6 For $n \geq 0$, we have $2^n = \sum_{k=0}^n \binom{n}{k}$

Theorem 15.8 For $n \in \mathbb{N}$, $k \geq 2$ and nonnegative integers $r_1, \dots, r_k$ such that $r_1 + \dots + r_k = n$, we have $\binom{n}{r_1, \dots, r_k} = \frac{n!}{r_1! r_2! \dots r_k!}$

Theorem 15.9 -The Binomial Theorem Let $n \in \mathbb{N}$ and $a, b \in \mathbb{C}$. Then $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$

Theorem 15.10 -The Multinomial Theorem Let $n \in \mathbb{N}$ and $x_1, \dots, x_k \in \mathbb{C}$. Then $(x_1 + \dots + x_k)^n = \sum \binom{n}{r_1, \dots, r_k} x_1^{r_1} \dots x_k^{r_k}$ where the sum is taken over all $r_i \in 0, 1, \dots, n$ such that $r_1 + \dots + r_k = n$

Theorem 15.11 -Inclusion Exclusion Principle Let $n \geq 2$ and suppose $A_1, A_2, \dots, A_n$ are finite sets. Let $N_1 = |A_1| + |A_2| + \dots + |A_n|$, $N_2 = |A_1 \cap A_2| + |A_1 \cap A_3| + |A_2 \cap A_3| + \dots + |A_{n-1} \cap A_n|$, and similarly, let $N_k = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} |A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}|$ be the sum of the sizes of the intersections of all collections of k different A_i 's, so that $N_n = |\bigcap_{k=1}^n A_k|$

Then it equals $\sum_{k=1}^n (-1)^{k+1} N_k$. In particular, for finite sets A,B,C $|A \cup B| = |A| + |B| - |A \cap B|$, and $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

Week 10: Lemma 16.1 The set S_n consists of exactly $n!$ permutations.

Lemma 16.2 The set S_n equipped with the composition rule $\circ$ has the following properties (1)Closure: for any f and g in S_n , one has $f \circ g \in S_n$, (2)Associativity: for any f, g , and h in S_n , one has $f \circ (g \circ h) = (f \circ g) \circ h$, (3)Identity element: there is unique permutation $\iota \in S_n$ such that $f \circ \iota = \iota \circ f = f$ for any $f \in S_n$ (and we call ι the identity permutation), (4)Inverse: for any $f \in S_n$, there is unique $f^{-1} \in S_n$ such that $f \circ f^{-1} = f^{-1} \circ f = \iota$.

Lemma 16.4 If $a_1, \dots, a_r$ are distinct numbers in $1, 2, \dots, n$ and $f = (a_1 a_2 \dots a_r)$ is a permutation, then the order of f is r .

Lemma 16.5 Let $f \in S_n$ have order m . If $f^k = \iota$ for some integer $k \in \mathbb{N}$, then m divides k .

Theorem 16.6 Any permutation $f \in S_n$ is a composite of disjoint cycles, that is, $f = f_1 \dots f_k$ for some $k \geq 1$ and disjoint cycles $f_1, \dots, f_k$.

Lemma 16.7 Let $f \in S_n$ and let m be its order. Write $f = \sigma_1 \sigma_2 \dots \sigma_s$ where $\sigma_1, \dots, \sigma_s$ are disjoint cycles of length $r_1, \dots, r_s$ respectively. Then $m = \text{lcm}(r_1, r_2, \dots, r_s)$, where lcm is the smallest positive integer that is divisible by r_i for each $i=1, \dots, s$.

Lemma 16.8 Every permutation in S_n is a product of cycles of length 2.

Lemma 16.9 Let $n \in \mathbb{N}$ (a) The sign of ι is 1 and any 2-cycle is -1 (b)or any f and g in S_n , we have $\text{sgn}(fg) = \text{sgn}(f)\text{sgn}(g)$. (c) the sign of a cycle of length r is $(-1)^{r-1}$ (d)For every permutation $f \in S_n$, let $f = \sigma_1 \sigma_2 \dots \sigma_s$ where $\sigma_1, \dots, \sigma_s$ are disjoint cycles of lengths $r_1, \dots, r_s$. Then $\text{sgn}(f) = (-1)^{r_1-1} (-1)^{r_2-1} (-1)^{r_3-1} \dots (-1)^{r_s-1}$ (e) If $f \in S_n$, then $\text{sgn}(f) = \text{sgn}(f^{-1})$

Corollary 16.10 Let f be a permutation in S_n (1) f is even if and only if it is a product of even number of cycles of order 2. (2) f is odd if and only if it is a product of odd number of cycles of order 2.